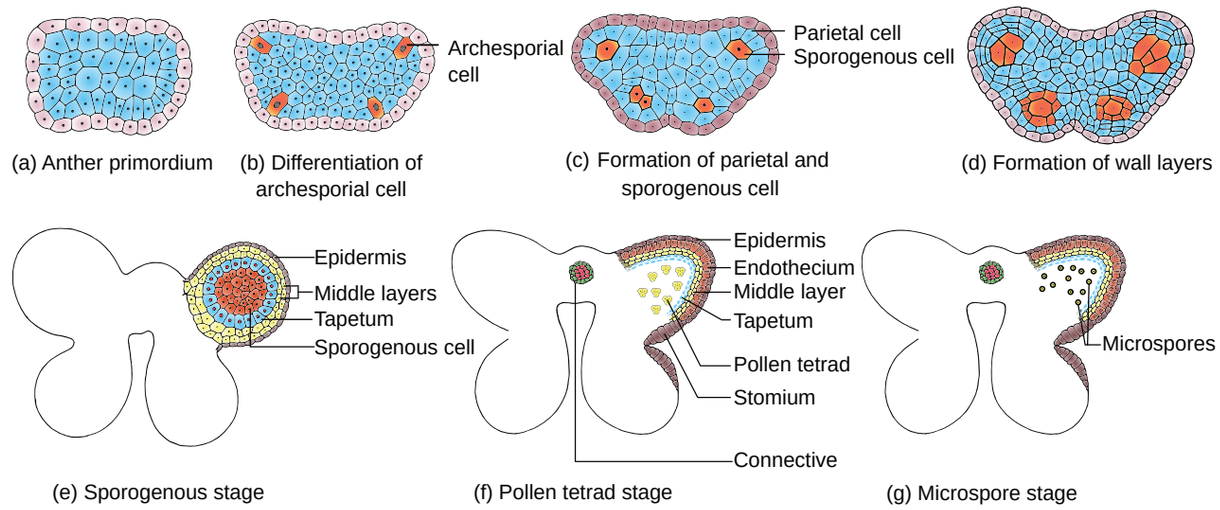


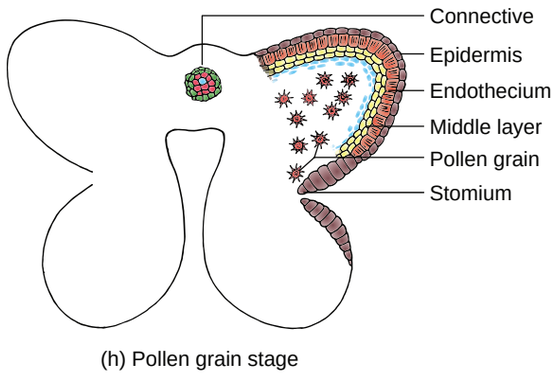




**Figure 1.4** Stages in the development of anther

attached to a clamp or clip like sticky structure called **corpusculum**. The filamentous or thread like part arising from each pollinium is called **retinaculum**. The whole structure looks like inverted letter 'Y' and is called **translator**

### T.S. of Mature anther

Transverse section of mature anther reveals the presence of anther cavity surrounded by an anther wall. It is bilobed, each lobe having 2 theca (dithecous). A typical anther is tetrasporangiate. The T.S. of Mature anther is given in Figure 1.5.

#### 1. Anther wall

The mature anther wall consists of the following layers **a. Epidermis b. Endothecium c. Middle layers d. Tapetum.**

**a. Epidermis**: It is single layered and protective in function. The cells undergo repeated anticlinal divisions to cope up with the rapidly enlarging internal tissues.

**Activity**

Collect buds and opened flowers of *Datura metel*. Dissect the stamens, separate the anthers and take thin transverse sections and observe the structure under the microscope. Record the various stages of anther development from your observations.

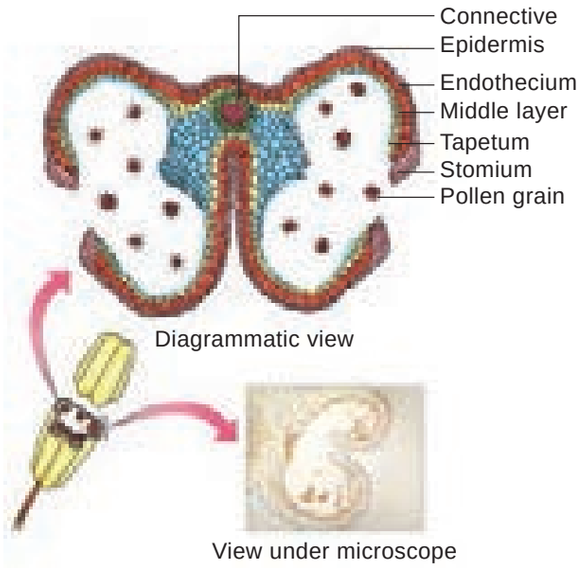



**Figure 1.5** T.S of Mature anther

**b.Endothecium**: It is generally a single layer of radially elongated cells found below the epidermis. The inner tangential wall develops bands (sometimes radial walls also) of α cellulose (sometimes also slightly lignified). The cells are **hygroscopic**. In the anthers of aquatic plants,



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